

Optimization with CasADi

Péter Kis, PhD

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Example #1 – Linear Programming

Farmer wants to grow **apples** and **bananas**.

	Area needed [m ² /ton]	Fertilizer needed [kg/ton]	Profit [USD/ton]
A	4000	60	700
B	3000	80	600

Farmer has a field of 100 000 m² and can legally use no more than 2000 kg fertilizer.
What to plant to maximize profit?

Example #1 – Formulation

m_a , m_b – mass of apple, and banana

Control variables,

$$w = [m_a, m_b]$$

the profit,

$$P(w) = 700m_a + 600m_b$$

the area used,

$$A(w) = 4000m_a + 3000m_b$$

fertilizer used,

$$F(w) = 60m_a + 80m_b$$

$$\max_w P(w) \rightarrow \min_w [-P(w)]$$

$$\text{s.t. } 0 \leq A(w) \leq 100000$$

$$0 \leq F(w) \leq 2000$$

Optional constraints:

$$0 \leq m_a, m_b$$

Example #1 – Solution in LibreOffice Calc

You go to menu Tools/Solver

	A	B	C	D	E	F	G	H
1		mass to plant (ton)						
2	Apple	14.2857142857143						
3	Banana	14.2857142857143						
4								
5	profit	18571.4285714286						
6								
7	s.t.							
8	area needed	100000	<=	100000				
9	fertilizer needed	2000	<=	2000				
10								
11								
12								
13								
14								
15								
16								

Solver

Target cell: **\$B\$5**

Optimize result to: ☒ Maximum
☐ Minimum
☐ Value of

By changing cells: **\$B\$2:\$B\$3**

Limiting Conditions

Cell reference	Operator	Value
\$B\$8	≤	\$D\$8
\$B\$9	≤	\$D\$9
	≤	
	≤	

Help Reset All Options... Close Solve

Example #1 – Solution in CasADi

```
from casadi import Opti

opti = Opti()

mass_apple = opti.variable()
mass_banana = opti.variable()
opti.minimize(-700 * mass_apple - 600 * mass_banana)

area_needed = 4000 * mass_apple + 3000 * mass_banana
fertilizer_needed = 60 * mass_apple + 80 * mass_banana

opti.subject_to(mass_apple >= 0)
opti.subject_to(mass_banana >= 0)

opti.subject_to(opti.bounded(0, area_needed, 100_000))
opti.subject_to(opti.bounded(0, fertilizer_needed, 2000))

opti.set_initial(mass_apple, 5)
opti.set_initial(mass_banana, 5)

opti.solver('ipopt')

solution = opti.solve()

print(f'Mass of apples: {solution.value(mass_apple):.2f} tons')
print(f'Mass of bananas: {solution.value(mass_banana):.2f} tons')
```

```
0 -6.5000000e+03 0.00e+00 8.21e-01 -1.0 0.00e+00 - 0.00e+00 0.00e+00 0
1 -9.3645636e+03 0.00e+00 1.32e+00 -1.0 4.41e+02 - 4.28e-01 1.00e+00f 1
2 -1.1401834e+04 0.00e+00 8.64e-01 -1.0 9.09e+02 - 5.86e-01 3.44e-01f 1
3 -1.7443302e+04 0.00e+00 3.55e-01 -1.0 1.18e+03 - 5.74e-01 7.30e-01f 1
4 -1.7539873e+04 0.00e+00 7.13e+00 -1.0 3.04e+01 - 9.09e-02 7.25e-01f 1
5 -1.8560419e+04 0.00e+00 7.28e-01 -1.0 5.23e+02 - 9.43e-02 9.10e-01f 1
6 -1.8570626e+04 0.00e+00 1.95e-02 -1.0 4.89e+00 - 1.00e+00 9.74e-01f 1
7 -1.8571389e+04 0.00e+00 2.83e-08 -2.5 1.19e-01 - 1.00e+00 1.00e+00f 1
8 -1.8571427e+04 0.00e+00 1.50e-09 -3.8 8.76e-03 - 1.00e+00 1.00e+00f 1
9 -1.8571429e+04 0.00e+00 1.84e-11 -5.7 4.85e-04 - 1.00e+00 1.00e+00f 1
iter objective inf_pr inf_du lg(mu) ||d|| lg(rg) alpha_du alpha_pr ls
10 -1.8571429e+04 0.00e+00 2.51e-14 -8.6 6.02e-06 - 1.00e+00 1.00e+00f 1

Number of Iterations.....: 10

                                (scaled)                                (unscaled)
Objective.....: -2.6530612326480496e+03 -1.8571428628536349e+04
Dual infeasibility.....: 2.5059035587193448e-14 1.7541324911035414e-13
Constraint violation.....: 0.0000000000000000e+00 0.0000000000000000e+00
Variable bound violation: 0.0000000000000000e+00 0.0000000000000000e+00
Complementarity.....: 2.5059135403446458e-09 1.7541394782412520e-08
Overall NLP error.....: 2.5059135403446458e-09 1.7541394782412520e-08

Number of objective function evaluations = 11
Number of objective gradient evaluations = 11
Number of equality constraint evaluations = 0
Number of inequality constraint evaluations = 11
Number of equality constraint Jacobian evaluations = 0
Number of inequality constraint Jacobian evaluations = 11
Number of Lagrangian Hessian evaluations = 10
Total seconds in IPOPT = 0.008

EXIT: Optimal Solution Found.
solver : t_proc (avg) t_wall (avg) n_eval
nlp_g | 522.00us ( 47.45us) 37.37us ( 3.40us) 11
nlp_g | 878.00us ( 79.82us) 63.89us ( 5.81us) 11
nlp_grad_f | 697.00us ( 58.08us) 50.20us ( 4.18us) 12
nlp_hess_l | 310.00us ( 31.00us) 21.80us ( 2.18us) 10
nlp_jac_g | 1.09ms ( 91.17us) 80.64us ( 6.72us) 12
total | 128.11ms (128.11ms) 9.45ms ( 9.45ms) 1
Mass of apples: 14.29 tons
Mass of bananas: 14.29 tons
```

Example #2 – Second order equation

$$P(w) = w^2 - 6w + 13$$

$$\min_w P(w)$$

$$\frac{dP(w)}{dw} = 2w - 6 = 0$$

$$w = 3, P(3) = 4$$

